COTW #51 Problem 2

Solutions by `@Frosty`, `@Theo` and `@BigTanuki`

Theorem. *For all positive integers n,*

$$\boxed{\sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{k} = \sum_{k=0}^{n} \binom{n}{k}^2 2^k.}$$

Proof (Algebraic). Recall that $\binom{n+k}{k} = \binom{n+k}{n}$, and recall that this is the coefficient of x^n in the expansion of $(x+1)^{n+k}$, for nonnegative integers n, k.

Define the magical function $\mathcal{C} : \mathbb{Z}[x] \times \mathbb{Z} \to \mathbb{Z}$, where $\mathcal{C}(f, n)$ is the coefficient of x^n in f, a polynomial of integer coefficients.

Write

$$\begin{aligned}
\sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{k} &= \sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{n} \\
&= \sum_{k=0}^{n} \binom{n}{k} \mathcal{C}\big((x+1)^{n+k}, n\big) \\
&= \mathcal{C}\left((1+x)^n \sum_{k=0}^{n} \binom{n}{k} (1+x)^k, n\right) && (\mathcal{C}(af, n) = a\mathcal{C}(f, n)) \\
&= \mathcal{C}((1+x)^n (2+x)^n, n).
\end{aligned}$$

Expanding this, we get

$$\begin{aligned}
\mathcal{C}((1+x)^n(2+x)^n, n) &= \mathcal{C}\left(\sum_{k=0}^{n} \binom{n}{k} x^k \sum_{j=0}^{n} \binom{n}{j} 2^{n-j} x^j, n\right) \\
&= \sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k} 2^k && (\text{Set } j = n-k) \\
&= \sum_{k=0}^{n} \binom{n}{k}^2 2^k,
\end{aligned}$$

and we are done. □

Remark. With help from `@BigTanuki`.

We now prove this result combinatorially.

Proof (Combinatorial). Suppose Theo has n K-pop songs and n J-pop songs saved. From these songs, suppose he first picks k K-pop songs and pools them together with all n J-pop songs. As there are $\binom{n}{k}$ ways to choose k K-pop songs and $\binom{n+k}{n}$ ways to form a playlist from this pool, there are $\binom{n}{k}\binom{n+k}{n}$ ways to form a playlist of length n. Hence, adding together the ways to pick such a playlist across all values of k, we get

$$\sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{n} = \sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{k}.$$

We now count this a different way. Suppose that Theo has k K-pop songs in his playlist of length n, and thus $n-k$ J-pop songs in his playlist. Immediately, there are $\binom{n}{k}\binom{n}{n-k} = \binom{n}{k}^2$ ways to choose k K-pop songs and $n-k$ J-pop songs. However, because all k K-pop songs in his playlist must come from a pool of k or more K-pop songs that he initially chooses, there are potentially up to $n-k$ more K-pop songs in the pool that remain unchosen. Hence, each way of forming a playlist has an additional factor of 2^{n-k} by counting the potential K-pop songs left in the pool. Summing over all k, we get

$$\sum_{k=0}^{n} \binom{n}{k}^2 2^{n-k} = \sum_{k=0}^{n} \binom{n}{k}^2 2^k,$$

and we are done. $\square$

Remark. With help from `@Theo`.

We provide another algebraic proof. Firstly, we prove a useful property of finite sums.

Lemma. *For all positive integers n, and $f : \mathbb{Z}^2 \to \mathbb{R}$,*

$$\boxed{\sum_{k=0}^{n} \sum_{j=0}^{k} f(k,j) = \sum_{j=0}^{n} \sum_{k=j}^{n} f(k,j).} \qquad (\phi)$$

Proof. Consider the following table of sum values for varying k and j.

k/j	0	1	2	$\cdots$	$n-1$	n
0	$f(0,0)$					
1	$f(1,0)$	$f(1,1)$				
2	$f(2,0)$	$f(2,1)$	$f(2,2)$			
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$		
$n-1$	$f(n-1,0)$	$f(n-1,1)$	$f(n-1,2)$	$\cdots$	$f(n-1,n-1)$	
n	$f(n,0)$	$f(n,1)$	$f(n,2)$	$\cdots$	$f(n,n-1)$	$f(n,n)$

Observe that in the left hand side, we take sums first across *rows*, and then down *columns*. Conversely, on the right hand side, we take sums first down *columns*, and then across *rows*. Yet as both these sums validly traverse the entire table, they must be equal. $\square$

Proof (Algebraic II).

$$\begin{aligned}
\sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{k} &= \sum_{k=0}^{n} \binom{n}{k} \sum_{j=0}^{k} \binom{n}{j}\binom{k}{j} \\
&= \sum_{k=0}^{n} \sum_{j=0}^{k} \binom{n}{k}\binom{n}{j}\binom{k}{j} \\
&= \sum_{j=0}^{n} \sum_{k=j}^{n} \binom{n}{k}\binom{n}{j}\binom{k}{j} && \text{(By lemma } \phi\text{)} \\
&= \sum_{j=0}^{n} \binom{n}{j} \sum_{k=j}^{n} \binom{n}{k}\binom{k}{j} \\
&= \sum_{j=0}^{n} \binom{n}{j}^2 \sum_{k=j}^{n} \binom{n-j}{k-j} \\
&= \sum_{j=0}^{n} \binom{n}{j}^2 2^{n-j} \\
&= \sum_{j=0}^{n} \binom{n}{j}^2 2^{j} \\
&= \sum_{k=0}^{n} \binom{n}{k}^2 2^{k},
\end{aligned}$$

and we are done. $\square$